

Project Initialization and Planning Phase

Date	July 3, 2026
Team ID	XXXXXX
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform credit card approval decisions using machine learning, boosting efficiency and accuracy. It tackles the inefficiencies of manual review, promising faster decisions, reduced risk exposure, and an improved applicant experience. Key features include multiple ML classification models and real-time approval prediction through a web application.

Project Overview	
Objective	The primary objective is to automate the credit card approval decision process by applying machine learning classification techniques, enabling faster, more consistent, and more accurate approval or rejection predictions.
Scope	The project covers data collection and preprocessing, training and comparing multiple classification algorithms, selecting the best-performing model, and deploying it through a Flask web application integrated with IBM Watson Machine Learning for real-time predictions.
Problem Statement	
Description	Manually reviewing thousands of credit card applications every day is slow, inconsistent, and prone to human error, leading to delayed decisions and inconsistent approval outcomes.
Impact	Automating the decision process will reduce processing time, minimize human bias and error, lower operational costs, and improve the overall applicant experience.
Proposed Solution	
Approach	Employing supervised machine learning classification algorithms to analyze historical applicant data and predict credit card approval outcomes, delivered through an easy-to-use web application.

Key Features	- Implementation and comparison of multiple ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost). - Feature engineering pipeline that converts multi-class payment status codes into clear binary risk labels. - Real-time approval/rejection prediction through a Flask web interface. - Cloud deployment via IBM Watson Machine Learning for scalable, real-time access.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (4 cores) — no GPU required for classical ML models
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	10 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, XGBoost, pandas, numpy, matplotlib
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle Credit Card Approval Prediction dataset, csv UCI Credit Approval dataset, 690, csv